

LECTURE NOTES MATH255

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1. LECTURE 15

In this class we introduce unique factorization domains and principal ideal domains.

1.1. Unique factorization domains. Let R be a commutative ring with unity. Recall that $a \in R$ divides $b \in R$ if there is some $c \in R$ such that $b = ac$ (we can write this as $a|b$). Moreover, recall that $u \in R$ is a unit if there is some $u' \in R$ such that $uu' = 1$.

Definition 1.1. We say that two elements $a, b \in R$ are associates in R if there is some unit $u \in R$ such that $b = au$.

Remark 1. Note that being associates is an equivalence relation: $a = 1 \cdot a$ and 1 is a unit, if $b = ua$ then $a = bu^{-1}$ where the multiplicative inverse of u , u^{-1} , is also a unit, and if $b = au$, $c = bu'$ then $c = auu'$ and uu' is a unit if u and u' are units (with multiplicative inverse $u^{-1}(u')^{-1}$).

Example 1.1. Let $R = \mathbb{Z}$, then the only units are ± 1 .

Example 1.2. Let $F[x]$ be the polynomial ring with F a field. The units in $F[x]$ are precisely the constant polynomials $a \in F \setminus \{0\}$.

Example 1.3. Let $R = \mathbb{Z}[\sqrt{7}]$. We consider the map $N : \mathbb{Z}[\sqrt{7}] \rightarrow \mathbb{Z}$ defined by:

$$(1.1) \quad N(a + b\sqrt{7}) = (a + b\sqrt{7})(a - b\sqrt{7}) = a^2 - 7b^2 \in \mathbb{Z}.$$

Notice that if $N(a + b\sqrt{7}) = 1$ then:

$$(1.2) \quad \frac{1}{a + b\sqrt{7}} = \frac{a - b\sqrt{7}}{(a + b\sqrt{7})(a - b\sqrt{7})} = \frac{a - b\sqrt{7}}{N(a + b\sqrt{7})} = a - b\sqrt{7} \in \mathbb{Z}[\sqrt{7}]$$

so if $N(a + b\sqrt{7}) = 1$ then $a + b\sqrt{7}$ is a unit in $\mathbb{Z}[\sqrt{7}]$. We can now find units in $\mathbb{Z}[\sqrt{7}]$ by calculating $N(a + b\sqrt{7})$:

$$(1.3) \quad N(8 + 3\sqrt{7}) = 64 - 7 \cdot 9 = 64 - 63 = 1.$$

That is, $u = 8 + 3\sqrt{7} \in \mathbb{Z}[\sqrt{7}]$ is a unit, and we can produce infinitely many units by considering $(8 + 3\sqrt{7})^n$ for $n \in \mathbb{Z}$.

Definition 1.2. Let R be an integral domain. A non-zero element $p \in R$ that is not a unit in R is irreducible (or an irreducible of R) if in any factorization $p = ab$ it either holds that $a \in R$ or $b \in R$ is a unit.

Date: February 18, 2026.

Remark 2. Note that if $p \in R$ is irreducible, then up is also irreducible for any unit u . So, all associates to p are irreducible as well if p is irreducible.

Example 1.4. If $R = F[x]$ with F a field then the irreducibles in $F[x]$ are precisely the irreducible polynomials in $F[x]$.

Example 1.5. If $R = \mathbb{Z}$ then the irreducibles in $\mathbb{Z}$ are precisely $\pm p$ with p prime.

Definition 1.3. Let R be an integral domain. We say that R is a unique factorization domain (UFD) if the following two conditions hold:

- (i) every element $a \in R$ that is not 0 and not a unit can be written as a product of irreducible elements in R ,
- (ii) if $a = p_1 \dots p_n = q_1 \dots q_m$ with $p_1, \dots, p_n, q_1, \dots, q_m$ irreducible in R then $n = m$ and after reordering the elements p_j and q_j are associates (i.e., there is a unit $u_j \in R$ such that $q_j = u_j p_j$).

Example 1.6. The unique factorization in $F[x]$, with F a field, proves that $F[x]$ is a UFD.

Example 1.7. The integers is a UFD, this follows since any integer $n \geq 2$ has a unique decomposition into primes.

Definition 1.4. An integral domain R is a principal ideal domain (PID) if every ideal $I \subset R$ is a principal ideal. That is, if $I \subset R$ is an ideal then $I = \langle a \rangle$ for some $a \in R$.

Example 1.8. If F is a field then we have shown that $F[x]$ is a principal ideal domain, since we proved that any ideal takes the form $I = \langle f(x) \rangle$ for some $f(x) \in F[x]$.

Example 1.9. Any ideal in $\mathbb{Z}$ is of the form $n\mathbb{Z} = \langle n \rangle$ (note that an ideal is an additive subgroup, and the only additive subgroups of $\mathbb{Z}$ are $n\mathbb{Z}$). It follows that $\mathbb{Z}$ is a PID.

1.2. PIDs are UFDs. We now start the proof that every PID is a UFD, we will finish the proof next class. The first step in proving that every PID is a UFD is to get a better understanding of the ideals in a PID.

Lemma 1.1. Let R be a commutative ring and let $I_1 \subset I_2 \subset I_3 \subset \dots$ be ideals in R , then:

$$(1.4) \quad I_\infty = \bigcup_{i=1}^{\infty} I_i$$

is an ideal in R .

Proof. Note that $0 \in I_1 \subset I_\infty$. Let $a, b \in I_\infty$ then $a \in I_i$ and $b \in I_j$ so $a, b \in I_{\max(i,j)}$. Since $I_{\max(i,j)}$ is an ideal so $-a, a + b \in I_{\max(i,j)} \subset I_\infty$. It follows that $(I_\infty, +)$ is an abelian group. It remains to show that if $a \in I_\infty$ and $b \in R$ then $ab \in I_\infty$. Let $a \in I_\infty$ then $a \in I_i$ for some i . Since I_i is an ideal we have $ab \in I_i$ for every $b \in R$, so $ab \in I_i \subset I_\infty$ and we conclude that I_∞ is an ideal. ■

Lemma 1.2 (Ascending chain condition). *Let R be a PID. If $I_1 \subset I_2 \subset I_3 \subset \dots$ are nested ideals in R then there is some $j < \infty$ such that $I_{j+k} = I_j$ for all $k \geq 0$.*

Proof. Let I_∞ be defined as in the previous lemma. Since R is a PID we find some $a \in R$ such that $I_\infty = \langle a \rangle$. Since $a \in I_\infty$ there is some j such that $a \in I_j$ and we conclude that $I_\infty = \langle a \rangle \subset I_j \subset I_\infty$ so $I_j = I_\infty$. If $k \geq 0$ then $I_j \subset I_{j+k} \subset I_\infty = I_j$ which implies $I_{j+k} = I_j$. ■

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